

Fiber-Optic Communication for Small Drones

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Focus: Computer Engineering, Embedded Systems, Optical Communications

Problem

Wireless communication is convenient, but that does not mean we should assume it will remain dependable in every environment where we want to use a drone. For instance, underground spaces, enclosed metallic structures, and industrial sites with substantial electromagnetic interference can make RF communication unreliable. Fiber optics offers a physical data link that does not depend on RF propagation. The practical question is whether we can use that link on a small drone without making the aircraft too heavy or difficult to operate.

Project Goal

We want to build and test a low-cost fiber-optic system for commands and telemetry, then investigate whether the fiber can be recovered and reused. The initial objective is to establish reliable communication and integrate it with a drone. Reusability is something we need to demonstrate, rather than assume from the outset.

Proposed System

Ground controller ↔ optical transceiver ↔ fiber-optic cable ↔ optical transceiver ↔ onboard controller ↔ flight controller

Commands would travel to the drone and telemetry would return through the same link. The flight controller would continue to stabilize the aircraft onboard. Video and other sensor data could be added later.

Research Questions

- How do latency, bandwidth, and packet reliability compare with a conventional wireless link under comparable test conditions?

- How much added weight, cable drag, and tension can a small drone tolerate while maintaining controlled flight?
- Can the fiber be deployed, recovered, and used again without damage? What would this require in cost, weight, and mechanical complexity?

Development Plan

Phase 1: Link testing. Establish bidirectional communication between two embedded systems and measure its performance.

Phase 2: Bench integration. Connect the fiber link to a flight controller and test commands and telemetry before attempting flight.

Phase 3: Controlled flight. Integrate the system with a small protected drone and evaluate how the cable affects flight.

Phase 4: Recovery and reuse. Investigate a motorized reel with tension sensing and fiber-length measurement to control payout and retrieval.

Potential Applications

Potential uses include tunnel and infrastructure inspection, industrial and hazardous-environment inspection, disaster response, and humanitarian or defense operations where a dependable RF link is unavailable.

Expected Outcome

The first outcome should be a working embedded communication prototype, measured performance and component costs, and documented hardware and software. A controlled flight demonstration would let us evaluate the integrated system. Further recovery tests would then help us determine whether repeated use is practical and what still needs to change.